

Adversarial Publication-Readiness Review and Revision Report

Correspondence and Logical Matrices: A Typed Boolean Operator Calculus

Review date: 27 September 2026

Object reviewed: the exact 31-page foundations manuscript supplied in `CM_LM_Publication_Readiness_Audit_2026-09-26.zip`, followed by the revised 32-page manuscript delivered with this report.

Review status: simulated independent specialist/referee lenses plus direct mathematical reconstruction, fresh literature spot-checking, executable verification, clean LaTeX build, and rendered-PDF inspection. These are not external human peer-review reports.

Executive conclusion

The revised manuscript is **ready for submission to an appropriate venue** as a foundations/methodology/representation paper. I found **no remaining central correctness, proof-completeness, reproducibility, or submission-hygiene blocker** after revision. The principal limitation is still **novelty and historical priority**: several constituent mechanisms have clear prior art or are standard Boolean/F2 constructions, so the current restrained claim - a typed, coherent integration of formula-valued LMs, numeric CMs, frame transport, agreement-bit pairing, and Boolean operator superposition - should be preserved.

The weighted readiness score under the requested rubric is **86.1 / 100**. That score is not a substitute for the priority caveat: the manuscript should not be sold as a major new Boolean algebra or as a collection of unprecedented elementary identities.

I would **not run another generic all-purpose review pass before submission**. The next useful pass should be venue-specific: template, length, audience, anonymization, MSC/keywords, supplement policy, and cover letter.

1. Had the paper already received hostile/adversarial review?

Yes, the supplied archive contains evidence of earlier adversarial work. Its nested provenance records refer to a `CM_Adversarial_Review_2026-09-18`, and the 26 September publication-readiness audit itself explicitly used adversarial/multiple-disciplinary perspectives and included a hostile-reading section and reviewer-feedback closure matrix.

However, two qualifications matter:

1. the 26 September audit explicitly says it was an AI-assisted independent analysis rather than a panel of separate human referees; and
2. that audit deliberately left the manuscript unchanged and reported its score against the unrepaired source/evidence package.

For this review I therefore treated the 31-page source as **unrepaired** and performed fresh specialist and hostile passes directly on that exact manuscript before editing it.

2. Fresh review protocol

The paper was attacked from eight distinct lenses:

1. **Boolean algebra / propositional logic specialist** - reconstructed the frame normal form, valuation, pairing, arbitrary outer Boolean superposition, and higher-arity lift.
2. **Boolean matrices / STP / history specialist** - checked dimensional conventions and exact antecedents in Boolean-matrix and semi-tensor-product literature.
3. **Finite linear algebra / tensor / rank specialist** - attacked flattening, matched-frame products, rank/separability, zero-rank and degenerate cases.
4. **Logic synthesis / symbolic computation specialist** - asked whether the architecture actually prevents an error or improves a derivation rather than merely renaming truth tables.
5. **Reproducibility / mathematical editorial specialist** - rebuilt the paper, executed the independent checker, inspected references and submission artifacts, and rendered the PDF.
6. **Hostile Referee A: novelty/priority** - assumed that every elementary ingredient was already known and tried to reduce the paper to “classical truth tables in new notation.”
7. **Hostile Referee B: correctness/specification** - searched for an explicit false statement, type error, counterexample, double action, degenerate failure, or illegal change of semantics.
8. **Handling-editor pass** - judged whether the revised paper has a coherent publishable center and what sort of venue can reasonably evaluate it.

The fresh finite checker was rerun after the mathematical revisions. It passed **2,022,648 explicitly counted assertions**. The final LaTeX source builds to **32 pages**, with no unresolved references/citations and no overfull or underfull box warnings. The PDF was rendered and visually inspected; targeted high-resolution checks covered the new worked example, the signed symbolic transport law, the related-work table, and the finite-verification appendix.

3. Strongest mathematical findings

3.1 What survived reconstruction

I did not find a counterexample to the central stated results when read under their declared assumptions. In particular, the following survived both proof reconstruction and the finite regression tests where finite checking is meaningful:

- binary formula-frame normal form and inverse;
- valuation to the correct signed numeric CM;
- agreement-bit semantics of logical pairing;
- arbitrary aligned Boolean operator superposition;
- preservation of superposition by valuation and pairing;
- higher-arity formula-valued LM lift;
- ordered tensor flattenings and bra-map-ket selection;
- matched-intermediate-frame XOR-AND matrix product;
- rank/conjunctive-XOR separability across a bipartition;
- signed input permutations and polarity changes after the transport law was made explicit;
- the stated ANF and finite spectral boundary claims.

The mathematics is strongest as a **coherent typed representation calculus**. It is not strongest as a source of deep new standalone Boolean theorems.

3.2 Real defects found by the hostile correctness pass

Three criticisms survived attack and were repaired:

- **STP vector orientation:** the manuscript used `vec([f])` as a row truth vector despite defining `vec` as a column vector. The revision uses the transpose where the row form is intended.
- **Signed symbolic transport:** the mask location was under-specified. The revision now states the exact symbolic permutation/polarity identity and warns that a polarity mask may be represented in the reference tuple or in the slot index, but not independently in both. A one-variable double-mask counterexample is included.
- **Degenerate conventions:** the zero-rank case and empty flattening blocks needed explicit empty-XOR/empty-product conventions. These are now stated.

These were genuine specification/local correctness issues, but none required withdrawing the central calculus.

4. Hostile novelty/priority attack

The strongest hostile reading remains serious and should be taken into account when choosing a venue:

- numeric CMs are reshaped truth tables;
- same-frame pointwise composition has direct prior art;
- signed input changes are closely related to standard NPN actions;
- the frame matrix has a close Boolean-space antecedent;
- arbitrary-arity truth tensors and flattenings are standard representation ideas;
- rank factorization over F2 is standard linear algebra;
- ANF and raw finite-field spectrum are standard coordinate/algebraic tools.

That attack is **substantially correct about the ingredients**. It does not make the manuscript mathematically false. The defensible contribution is the integration of those ingredients into one explicit symbolic/numeric, frame-aware architecture with formula-valued polarity objects, agreement-bit pairing, typed transport, and a coherence theorem linking the levels.

The revision responds by making that synthesis claim explicit, rather than attempting to inflate elementary ingredients into novelty claims.

5. Fresh literature and priority spot-check

The most important current comparisons were checked directly against primary or author-hosted sources:

- **Cheng, Zhao, Xu (2011), “Matrix Approach to Boolean Calculus”**: Proposition 3.3 states the same-frame entrywise truth-table law for an arbitrary binary logical operator. The revised paper now treats the corresponding CM law as prior numerical structure rather than as a new result.
- **Gudder and Latremoliere (2009), “Boolean Inner Product Spaces and Boolean Matrices”**: Example 2.13 constructs an orthonormal basis by cyclically shifting a stochastic vector. In dimension two, the mechanism gives the complementary-row Boolean frame used by the LM frame matrix. The algebraic setting is not identical to the present XOR-AND/formula-valued architecture, but the overlap is close enough that a direct comparison is required and is now present.
- **Toffano (2020), “Eigenlogic in the Spirit of George Boole”**: the manuscript now cites the published version of record rather than relying on a fragile version-specific historical statement.
- **Zhao, Gao, Cheng, “Semi-Tensor Product Approach to Boolean Functions”**: the identifiable author-hosted preprint is cited without inventing a publication date.
- **Usturali, Chamon, Ruckenstein, Mucciolo (2025), “A Matrix Product State Representation of Boolean Functions”**: this introduces a Binary Matrix Product normal form aimed at compressed/TT-

MPS-like representation. The revised manuscript distinguishes that objective from its exact, uncompressed truth-tensor architecture.

Targeted searches did **not** surface an exact antecedent for the complete formula-valued LM + agreement pairing + signed frame typing + coherence package. That is useful evidence for differentiation, **not proof of historical firstness**. This is why novelty/priority remains the lowest major score.

6. Revision actions taken

The revised manuscript incorporates the reasonable and technically supported reviewer feedback. Major changes include:

- sharpened title: **A Typed Boolean Operator Calculus**;
- contribution statement rewritten around typed synthesis and exact claim boundaries;
- direct Cheng and Gudder-Latremoliere comparisons;
- corrected STP row/column convention;
- explicit signed symbolic transport identity and double-mask warning;
- expanded higher-arity pairing proof;
- explicit zero-rank empty-XOR and degenerate flattening conventions;
- noncontiguous ordered-flattening example;
- complete end-to-end frame-alignment example showing a concrete wrong result before transport and the corrected result after transport;
- clearer separation of pointwise Boolean superposition from XOR-AND matrix multiplication;
- reduced phase/quantum-adjacent material;
- renamed spectral appendix;
- repaired Toffano, Zhao/Gao/Cheng, Mizraji, and related bibliography details;
- added the 2025 Binary Matrix Product comparison;
- replaced the unsupported historical finite-check provenance in the paper with the new independently executable verification supplement;
- removed audit/process wording from the submission manuscript and cleaned PDF metadata/layout.

A line-by-line unified diff and a more detailed revision changelog are included in the bundle.

7. Publication-readiness scorecard

Category	Weight	Score / 10	Confidence
Mathematical correctness	20%	9.3	High
Completeness of proofs/results	8%	9.1	High
Model and assumption discipline	8%	9.3	High
Novelty and priority	12%	6.2	Moderate
Scientific significance	8%	7.4	Moderate
Literature coverage and positioning	8%	8.7	Moderate-High
Reproducibility	7%	9.2	High
Internal consistency	6%	9.4	High
Resolution of prior reviewer concerns	5%	9.4	High
Exposition and readability	5%	8.7	High
Scope and structural focus	4%	8.8	High
Evidence and citation quality	4%	8.7	Moderate-High
Submission hygiene	3%	9.1	High
Venue/form fit	2%	8.0	Moderate

Weighted total: 86.1 / 100.

Scorecard notes

- **Mathematical correctness:** No central theorem counterexample was found; the genuine local specification/type defects were repaired.
- **Completeness of proofs/results:** Higher-arity selection and degenerate rank/block conventions are now explicit.
- **Model and assumption discipline:** Formula/numeric levels, frame transport, pointwise superposition, XOR-AND multiplication, and spectral boundaries are cleanly separated.
- **Novelty and priority:** Many ingredients have direct prior art. No exact integrated duplicate surfaced in targeted search, but firstness is not certified.
- **Scientific significance:** The typed synthesis is useful and the frame-error example demonstrates value; broader impact remains application-dependent.
- **Literature coverage and positioning:** Direct comparisons now cover the closest numerical/frame antecedents, STP, Eigenlogic/NPN, and a modern BMP representation.
- **Reproducibility:** A fresh self-contained checker passes 2,022,648 counted assertions; unsupported historical check-count provenance is no longer used.
- **Internal consistency:** STP orientation and signed symbolic transport are now explicit and consistent.
- **Resolution of prior reviewer concerns:** Foundation-specific P1/P2 issues in the supplied audit were addressed or correctly scoped out.

- **Exposition and readability:** The running example and tighter framing improve the narrative; the paper remains mathematically dense.
- **Scope and structural focus:** The foundations paper now owns the typed calculus; performance and phase extensions are clearly separated.
- **Evidence and citation quality:** Primary-source positioning is much stronger; some historical chronology is inherently uncertain.
- **Submission hygiene:** The final 32-page build has clean references/boxes, verified rendering, and cleaned title/metadata.
- **Venue/form fit:** It is a good methodological/foundational representation paper, but a weaker fit for venues demanding one deep new theorem.

The numerical score does not override novelty/priority. The manuscript is submission-ready **because there is no remaining serious correctness or evidence blocker and because its claims have been calibrated to the evidence**, not because every category is near 10.

8. Submission decision

Decision: READY FOR SUBMISSION, WITH VENUE-MATCH CAVEAT

I would submit the revised manuscript now to a venue that is receptive to one or more of the following:

- Boolean-function or logical representations;
- algebraic/symbolic methods in logic or computation;
- finite Boolean matrix methods;
- formal representation/specification calculi;
- methodological mathematical notes where integration and exact typing are legitimate contributions.

I would **not** submit it with a cover letter or abstract that sells it as a major new Boolean algebra, a new physical/quantum formalism, a compressed Boolean representation, or a demonstrated computational speedup. I also would not choose a venue whose acceptance bar is dominated by a single deep theorem unless new mathematics is added first.

There is no strong reason for another generic adversarial pass before submission. The highest-value next step is a **target-journal pass** that adjusts framing and length to the chosen audience without reopening settled mathematics.

9. Remaining non-blocking items before actual upload

These depend on the target venue rather than the mathematical manuscript:

- author affiliation and contact line, if required;

- ORCID and funding/conflict/data statements, if required;
 - anonymization for double-blind review, if required;
 - MSC/ACM classification codes, if requested;
 - journal template and page/abstract limits;
 - naming and citation of the checker/supplement according to the journal’s data policy;
 - a cover letter that describes the contribution as a typed synthesis and names the closest antecedents explicitly.
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10. Delivered artifacts

The final bundle contains:

- revised LaTeX source and 32-page PDF;
- the pre-review source/PDF snapshot;
- eight separate reviewer/hostile/editorial memos;
- revision changelog and full unified source diff;
- readiness scorecard in CSV and JSON;
- independent checker, machine-readable verification results, compact CM atlas, and boundary counterexamples;
- SHA-256 manifest for the delivered bundle.